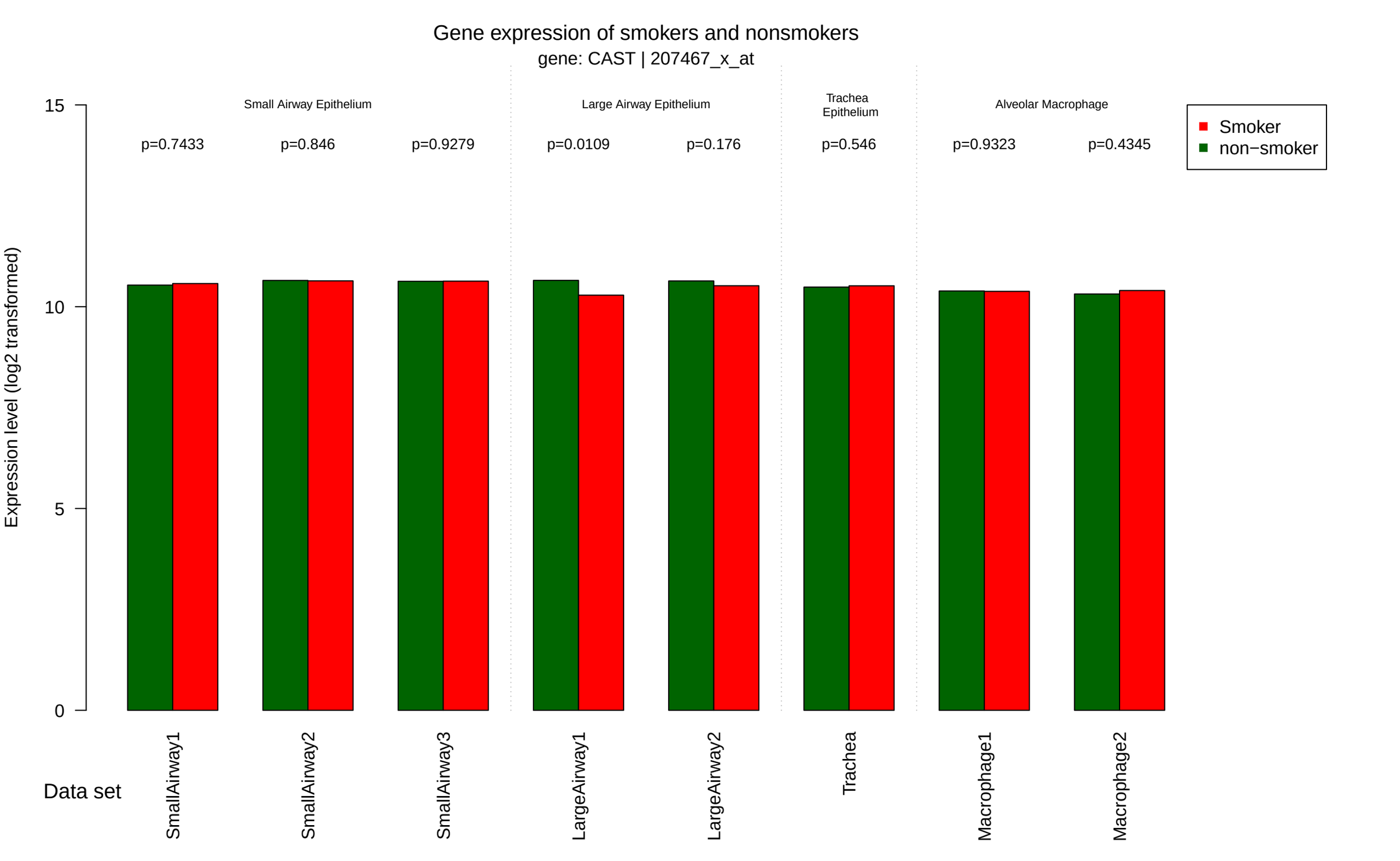
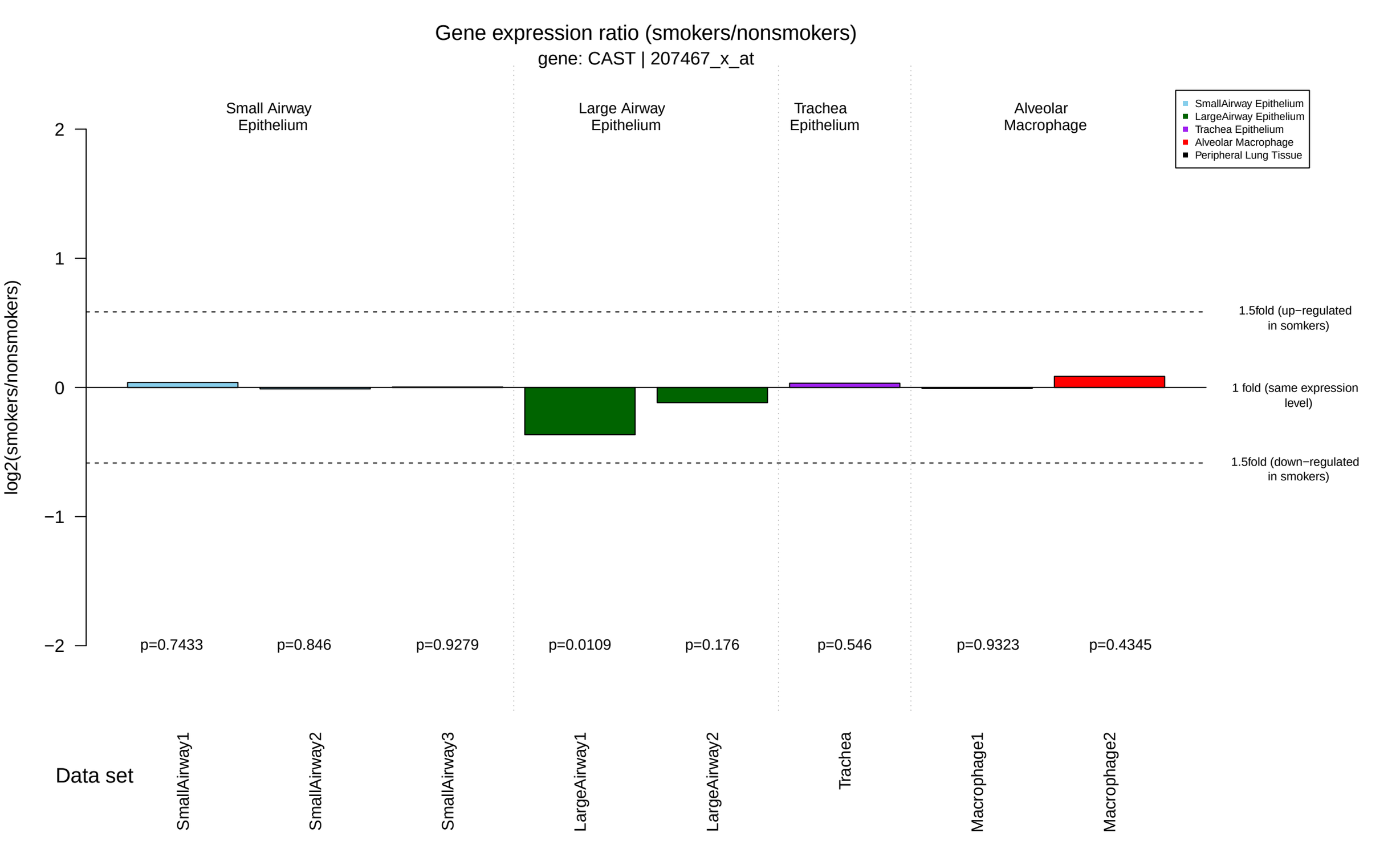
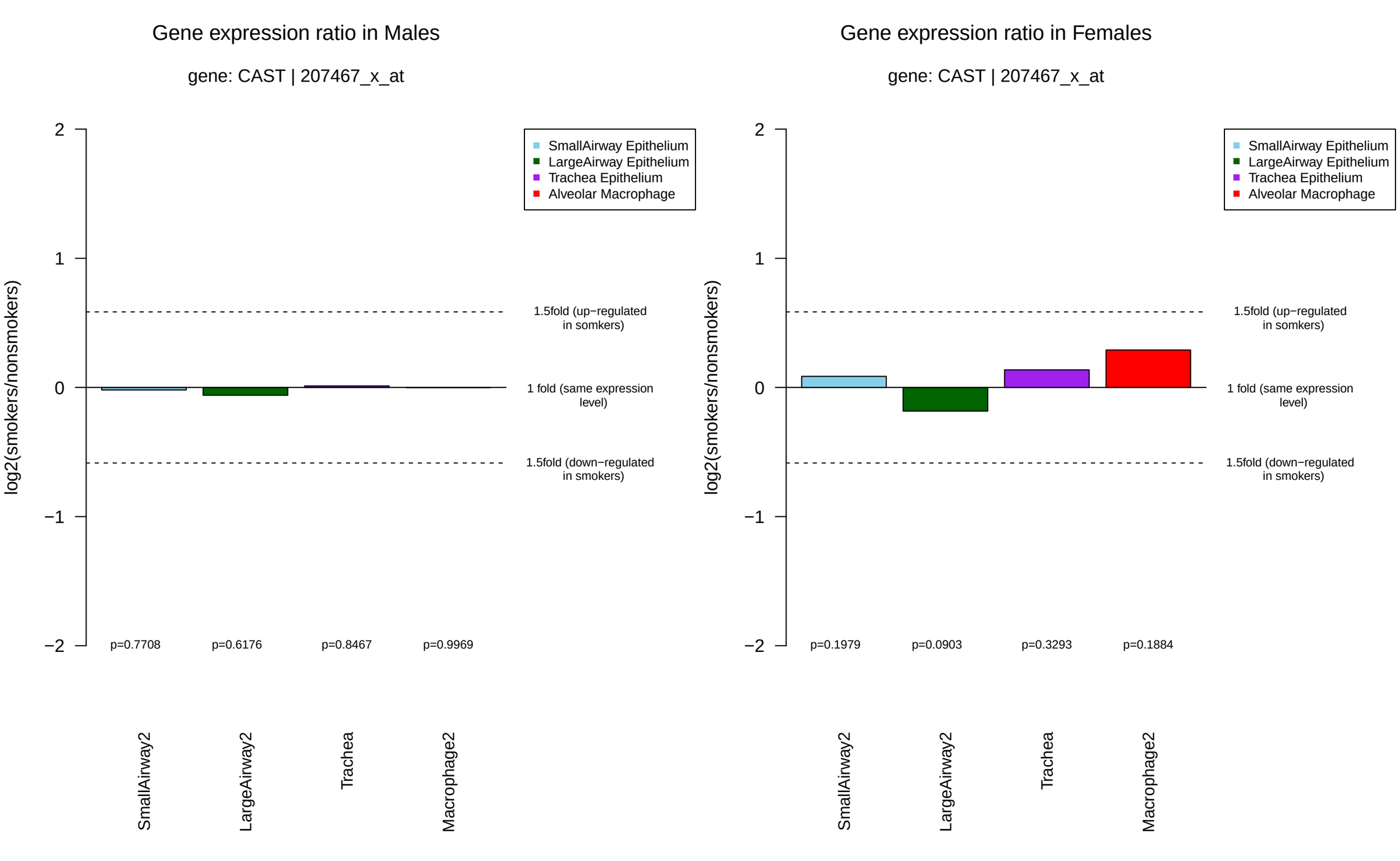
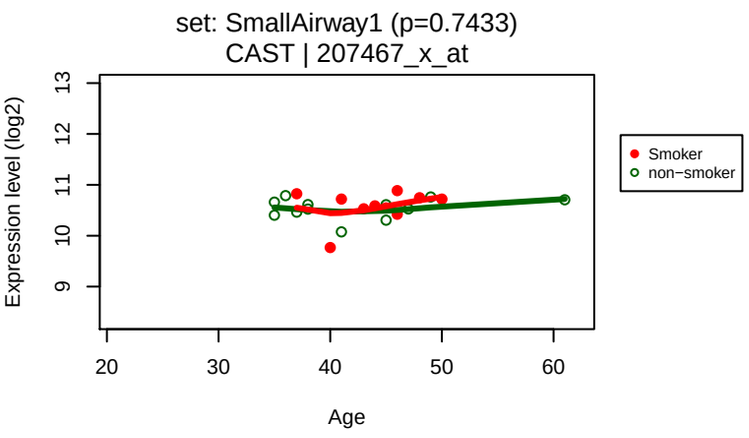
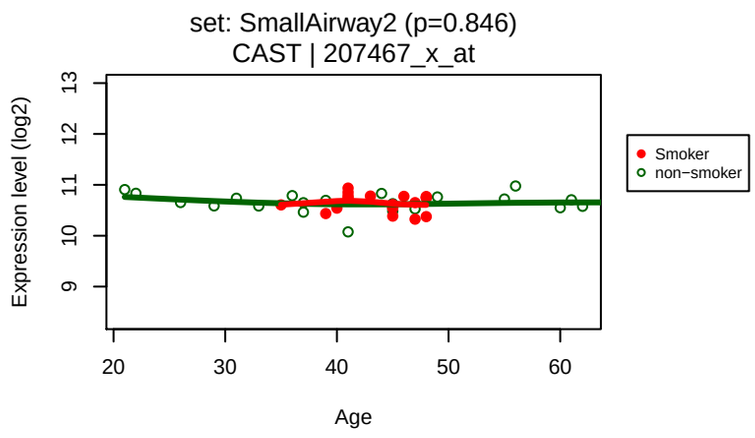
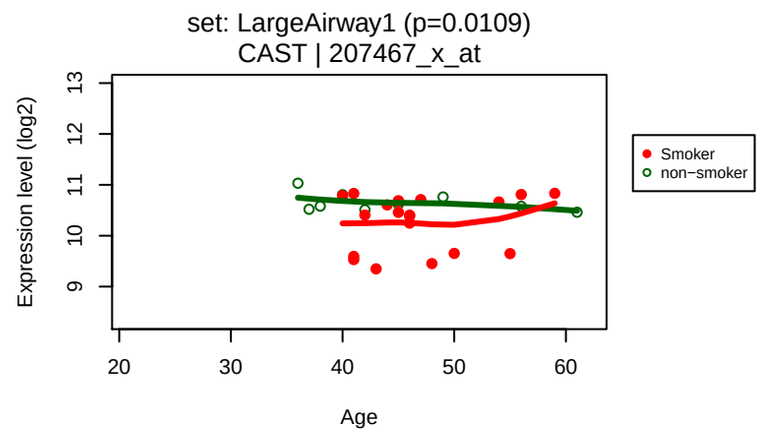

Gene expression of smokers and nonsmokers
gene: CAST | 207467_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7433
p=0.846
p=0.9279
p=0.0109
p=0.176
p=0.546
p=0.9323
p=0.4345
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

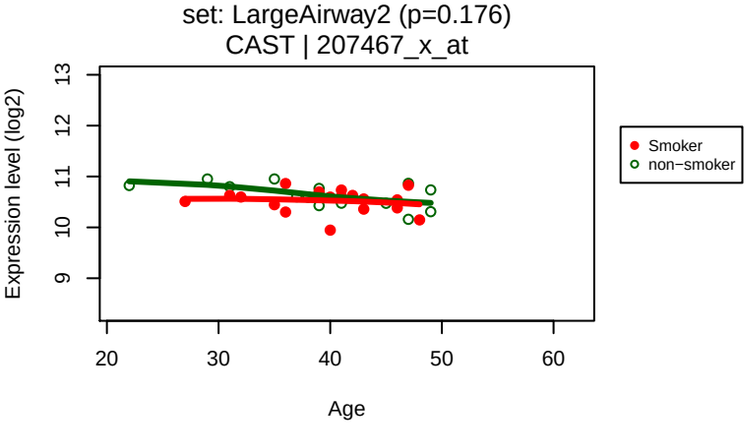

Gene expression ratio (smokers/nonsmokers)
gene: CAST | 207467_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7433
p=0.846
p=0.9279
p=0.0109
p=0.176
p=0.546
p=0.9323
p=0.4345
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

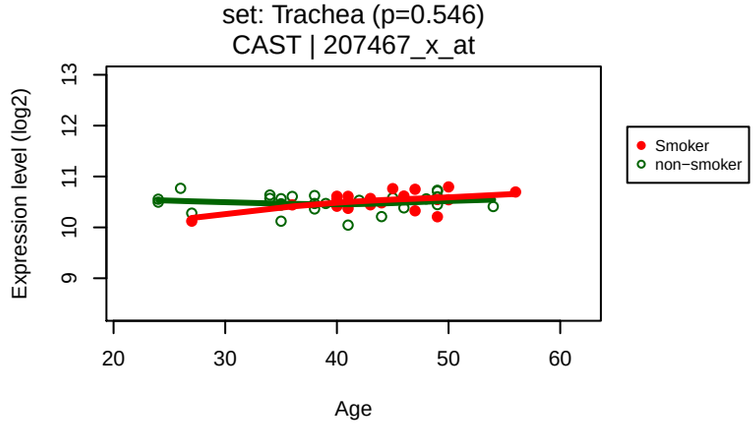

Gene expression ratio in Males
gene: CAST | 207467_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7708
p=0.6176
p=0.8467
p=0.9969
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CAST | 207467_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1979
p=0.0903
p=0.3293
p=0.1884
SmallAirway2
LargeAirway2
Trachea
Macrophage2

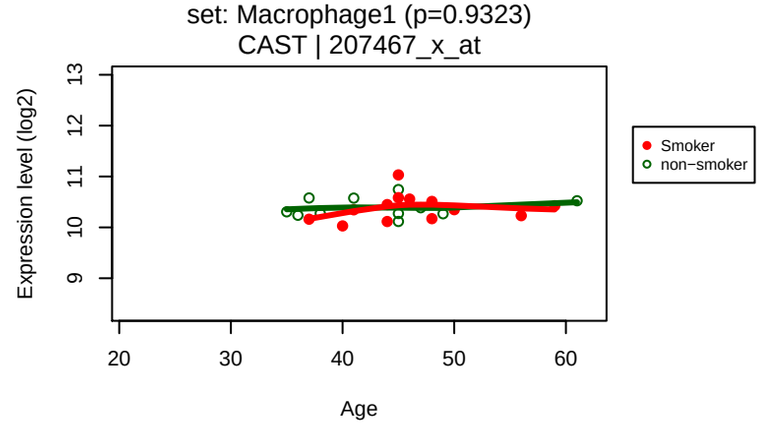

set: SmallAirway1 (p=0.7433)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker

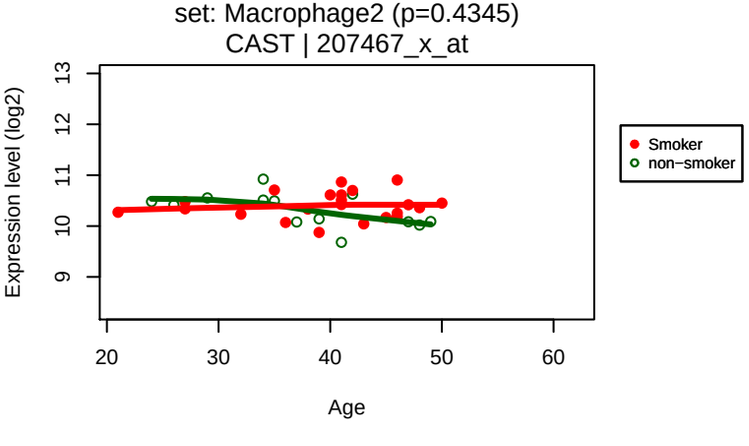

set: SmallAirway2 (p=0.846)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker

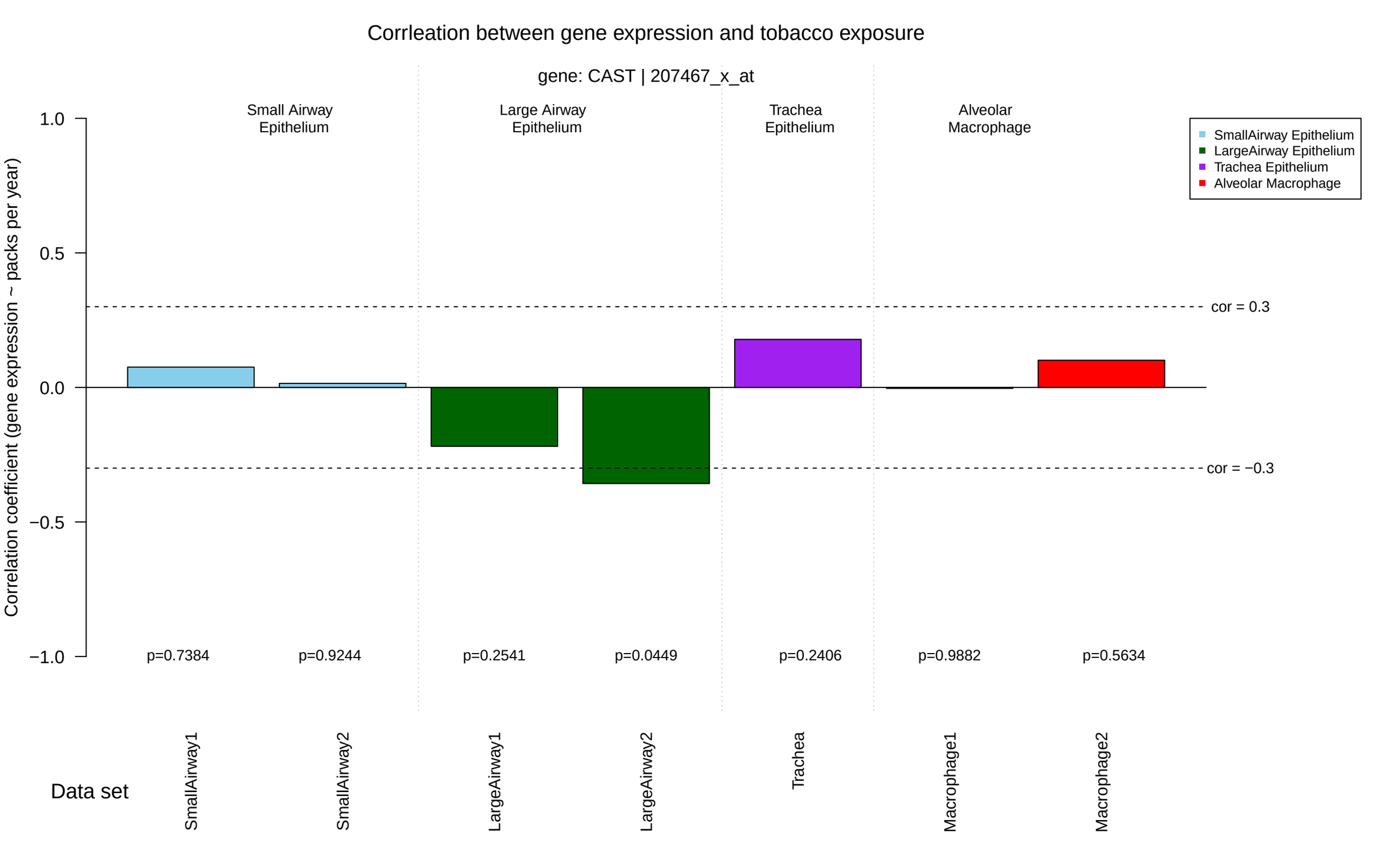

set: LargeAirway1 (p=0.0109)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.176)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.546)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9323)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4345)
CAST | 207467_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CAST | 207467_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7384
p=0.9244
p=0.2541
p=0.0449
p=0.2406
p=0.9882
p=0.5634
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2